Final Report

CARIS LIFE SCIENCES

Patient

Name: BINDAL, YASHDEEP
Date of Birth: [redacted]
Sex: Male
Case Number: TN25-112103
Diagnosis: Adenocarcinoma, NOS

Specimen Information

Primary Tumor Site: Pancreas, NOS
Specimen Site: Liver
Specimen ID: [redacted]
Specimen Collected: 15-Jan-2025
Test Report Date: 22-Feb-2025

Ordered By

[redacted]

Results with Therapy Associations

BIOMARKER	METHOD	ANALYTE	RESULT	THERAPY ASSOCIATION		BIOMARKER LEVEL*
NRG1	**Seq**	**RNA-Tumor**	**Pathogenic Fusion**	**BENEFIT**	**zenocutuzumab**	**Level 2**

* Biomarker reporting classification - Level 1 – MI Cancer Seek™ or other companion diagnostic (CDx) performed as part of professional services; Level 2 – Strong evidence of clinical significance or is endorsed by standard clinical guidelines; Level 3 – Potential clinical significance. Bolded benefit therapies, if present, highlight the most clinically significant findings.

Important Note

This content is provided as part of the professional services. For MI Cancer Seek™ results, see CDx Associated Findings.

A CDH1::NRG1 fusion was detected by RNA sequencing. This fusion is in-frame and similar to other reported fusions involving NRG1 (PMID: 25531467).

Zenocutuzumab-zbco, a HER2/HER3 bi-specific antibody, is now FDA-approved for advanced, unresectable or metastatic non-small cell lung cancer (NSCLC) and pancreatic adenocarcinoma patients harboring NRG1+ (neuregulin 1) gene fusion with disease progression on or after prior systemic therapy. This accelerated approval is based on findings from the eNRGy trial (NCT02912949).

Cancer-Type Relevant Biomarkers

Biomarker	Method	Analyte	Result
KRAS	**Seq**	**DNA-Tumor**	**Mutation Not Detected**
BRAF	Seq	DNA-Tumor	Mutation Not Detected
		RNA-Tumor	Fusion Not Detected
MSI	Seq	DNA-Tumor	Stable
NTRK1/2/3	Seq	RNA-Tumor	Fusion Not Detected
RET	Seq	RNA-Tumor	Fusion Not Detected
Tumor Mutational Burden	Seq	DNA-Tumor	Low, 3 mut/Mb
ALK	Seq	RNA-Tumor	Fusion Not Detected
ATM	Seq	DNA-Tumor	Mutation Not Detected

Biomarker	Method	Analyte	Result
BRCA1	Seq	DNA-Tumor	Mutation Not Detected
BRCA2	Seq	DNA-Tumor	Mutation Not Detected
ERBB2 (Her2/Neu)	CNA-Seq	DNA-Tumor	Amplification Not Detected
	IHC	Protein	Negative | Score 0
FGFR2	Seq	RNA-Tumor	Fusion Not Detected
MTAP	CNA-Seq	DNA-Tumor	Deletion Not Detected
PALB2	Seq	DNA-Tumor	Mutation Not Detected
PD-L1 (SP142)	IHC	Protein	Negative | 0%
ROS1	Seq	RNA-Tumor	Fusion Not Detected

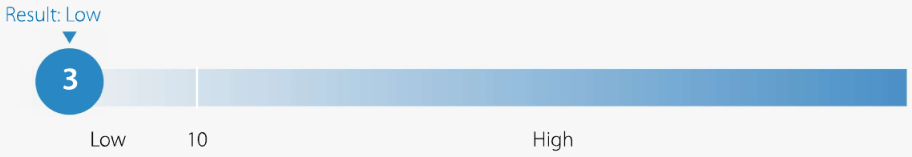
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The selection of any, all, or none of the matched therapies resides solely with the discretion of the treating physician. Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all available information concerning the patient's condition, the FDA prescribing information for any therapeutic, and in accordance with the applicable standard of care. Whether or not a particular patient will benefit from a selected therapy is based on many factors and can vary significantly. All trademarks and registered trademarks are the property of their respective owners.

Cancer-Type Relevant Biomarkers (continued)

Biomarker	Method	Analyte	Result
SMAD4	Seq	DNA-Tumor	Mutation Not Detected

Genomic Signatures

Biomarker	Method	Analyte	Result
Microsatellite Instability (MSI)	Seq	DNA-Tumor	Stable
Tumor Mutational Burden (TMB)	Seq	DNA-Tumor	
Genomic Loss of Heterozygosity (LOH)	Seq	DNA-Tumor	Low - 8% of tested genomic segments exhibited LOH (assay threshold is ≥ 16%)

Genes Tested with Pathogenic or Likely Pathogenic Alterations

Gene	Method	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %
NRG1	Seq	RNA-Tumor	Pathogenic Fusion	CDH1-NRG1	6	-	-
TERT	Seq	DNA-Tumor	Pathogenic Variant	-	0	c.-124C>T	33
TGFBR2	Seq	DNA-Tumor	Pathogenic Variant	p.N490fs	6	c.1469dupA	38

Unclassified alterations for DNA and RNA sequencing can be found in the MI Portal.
Formal nucleotide nomenclature and gene reference sequences can be found in the Appendix of this report.
Variants of Uncertain Significance can be found in the MI Portal.

[redacted]
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Predicted Human Leukocyte Antigen (HLA) Genotype Results

The predicted HLA genotype results are summarized in the table below. Please note that HLA typing via tumor tissue sequencing is not a substitute for histocompatibility testing performed using peripheral blood. It is recommended to confirm the patient's HLA type with an appropriate assay.

Gene	Method	Analyte	Genotype
MHC CLASS I			
HLA-A	Seq	DNA-Tumor	A*01:01, A*33:03
HLA-B	Seq	DNA-Tumor	B*08:01, B*15:08
HLA-C	Seq	DNA-Tumor	C*01:02, C*07:02

HLA genotypes with only one allele are either homozygous or have loss-of-heterozygosity at that position.

Immunohistochemistry Results

Biomarker	Result	Biomarker	Result
ERBB2 (Her2/Neu)	Negative | Score 0	PD-L1 (SP142)	Negative | 0%

Genes Tested with Indeterminate Results by Tumor DNA Sequencing

ATP6AP2	COL2A1	CUL3	KIF1B	PLCB4	PRDM6	PRKD1	RASA1	REST	SMARCA2	STAG2	

Genes in this table were ruled indeterminate due to low coverage for some or all exons.

Genes Tested with Intermediate CNA Results by Tumor DNA Sequencing

AKT3	DDR2	MCL1	NTRK1	RAC1							

The results in this report were curated to represent biomarkers most relevant for the submitted cancer type. These include results important for therapeutic decision-making, as well as notable alterations in other biomarkers known to be involved in oncogenesis. Additional results, including genes with normal findings, variants of uncertain significance, or unclassified alterations can be found in the MI Portal at **miportal.carismolecularintelligence.com.** If you do not have an MI Portal account, or need assistance accessing it, please contact Caris Customer [redacted]

[redacted]
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Notes of Significance

SEE APPENDIX FOR DETAILS

CDx reports are generated through automated bioinformatics processing of a pre-defined list of genes and variants. Pathogenic or likely pathogenic variants identified outside the FDA-approved list will not be included in the CDx report. In addition, downgrading of variants approved by board-eligible/board-certified geneticists and pathologists may have occurred through manual variant annotation carried out by Caris' professional services. To ensure optimal coverage, multiple sequencing runs may be performed for a given case. This may account for some discrepancies observed between CDx reports and professional services.

A CDH1::NRG1 fusion was detected by RNA sequencing. This fusion is in-frame and similar to other reported fusions involving NRG1 (PMID: 25531467). Ekaterina Bogdanova, PhD, FACMG, 02/22/2025

Clinical Trials Connector ™ opportunities based on biomarker expression: 2 Targeted Therapies. See page 5 for details.

Specimen Information

Specimen ID: [redacted]

Specimen Collected: 01/15/2025

Specimen Received: 02/14/2025

Testing Initiated: 02/17/2025

Test Ordered*: MI Profile™ (MI Cancer Seek™ + IHCs and Other Tests by Tumor Type)
* If the submitted specimen is inadequate, only a subset of the ordered testing may be reported.

Gross Description: 1 (A) Paraffin Block - Client [redacted] from Stanford Health Care - Stanford University Medical Center, Stanford, CA, with the corresponding pathology report labeled [redacted]

Dissection Information: A laboratory technician harvested targeted tissues for extraction from the marked areas using a dissection microscope.

H&E Tumor Assessment:
Professional Component Performed: - - Location 67, Chapel Hill, NC. [redacted] Lab Director.

Electronic Signature
[redacted]
02/17/2025

[redacted]
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Clinical Trials Connector™

The Clinical Trials Connector lists agents that are matched to available clinical trials according to biomarker status. In some instances, older-generation agents may still be relevant in the context of new combination strategies and, therefore, will still appear on this report.

Therapeutic agents listed below may or may not be currently FDA approved for the tumor type tested.

Please see https://clinicaltrials.gov/ for more information.

TARGETED THERAPIES (2)				
Drug Class	**Biomarker**	**Method**	**Analyte**	**Investigational Agent(s)**
Multi-HER-targeted therapy (2)	NRG1	RNA-Seq	RNA-Tumor	afatinib

( ) = represents the total number of clinical trials identified by the Clinical Trials Connector for the provided drug class or table.

The Clinical Trials Connector may include trials that enroll patients with additional screening of molecular alterations. In some instances, only specific gene variants may be eligible.

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Disclaimer

Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all available information concerning the patient's condition, prescribing information for any therapeutic, and in accordance with the applicable standard of care. Drug associations provided in this report do not guarantee that any particular agent will be effective for the treatment of any patient or for any particular condition. Caris Life Sciences® expressly disclaims and makes no representation or warranty whatsoever relating, directly or indirectly, to the performance of services, including any information provided and/or conclusions drawn from therapies that are included or omitted from this report. Whether or not a particular patient will benefit from a selected therapy is based on many factors and can vary significantly. The selection of therapy, if any, resides solely in the discretion of the treating physician and the tests should not be considered a companion diagnostic.

Caris MPI, Inc. d/b/a Caris Life Sciences is certified under the Clinical Laboratory Improvement Amendments (CLIA) as qualified to perform high complexity clinical laboratory testing, including all Caris molecular profiling assays. Individual assays that are available through Caris molecular profiling include both Laboratory Developed Tests (LDT) and U.S. Food and Drug Administration (FDA) approved or cleared tests. In addition, certain tests have been CE-marked as a general IVD under the In Vitro Diagnostic Directive (IVDD) 98/79/EC. Offered LDTs were developed and their performance characteristics determined by Caris. Certain tests have not been cleared or approved by the FDA. Caris LDTs are used for clinical purposes. They are not investigational or for research. Caris' CLIA certification number is located at the bottom of each page of this report.

The information presented in the Clinical Trials Connector™ section of this report, if applicable, is compiled from sources believed to be reliable and current. However, the accuracy and completeness of the information provided herein cannot be guaranteed. The clinical trials information present in the biomarker description was compiled from www.clinicaltrials.gov. The contents are to be used only as a guide, and health care providers should employ their best comprehensive judgment in interpreting this information for a particular patient. Specific eligibility criteria for each clinical trial should be reviewed as additional inclusion criteria may apply.

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Caris molecular testing is subject to Caris' intellectual property. Patent www.CarisLifeSciences.com/ip.

Professional Component Performed: - - Location 8, Phoenix, AZ. [redacted] Lab Director.

[redacted]

Electronic Signature

[redacted]

22-Feb-2025

[redacted]

MI Cancer Seek™

CARIS | MI CANCER SEEK

Patient

Name: BINDAL, YASHDEEP
Date of Birth: [redacted]
Sex: Male
Case Number: TN25-112103
CDx Report Generated: 02/22/2025

Specimen Information

Diagnosis and Tumor Type:
Adenocarcinoma, NOS, Pancreas, NOS
Specimen Site: Liver
Specimen ID: [redacted]
Specimen Type: Formalin-fixed paraffin-embedded
Specimen Collected: 01/15/2025

Ordered By

[redacted]

CDx Associated Findings

Genomic Findings Detected	FDA-approved Therapeutic Options
Findings did not yield any biomarker results with Companion Diagnostic (CDx) Claims.	
Microsatellite Instability (MSI)	Not MSI-H

Tumor Profiling Results

MI Cancer Seek is FDA-approved to provide tumor mutation profiling results for previously diagnosed oncology patients with solid tumors.

Other Alterations and Biomarkers Identified

Results reported in this section are not prescriptive or conclusive for labeled use of any specific therapeutic product.
Confirmation of tumor mutation status using an FDA-approved CDx test is needed for therapeutic use.

Tumor Mutational Burden (TMB)	3 mut/Mb
TERT	c.-124C>T

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MI Cancer Seek™

Intended Use:

MI Cancer Seek is a next-generation sequencing (NGS) based in vitro diagnostic (IVD) device using total nucleic acid (TNA) isolated from formalin-fixed paraffin-embedded (FFPE) tumor tissue specimens for the detection of single nucleotide variants (SNVs) and insertions and deletions (indels) in 228 genes, microsatellite instability (MSI), tumor mutational burden (TMB) in patients with previously diagnosed solid tumors, and copy number amplification (CNA) in one gene in patients with breast cancer.

MI Cancer Seek is intended as a companion diagnostic to identify patients who may benefit from treatment with the targeted therapies listed in Table 1 below, in accordance with the approved therapeutic product labeling.

Additionally, MI Cancer Seek is intended to provide tumor mutational profiling to be used by qualified healthcare professionals in accordance with professional oncology guidelines for cancer patients with previously diagnosed solid malignant neoplasms. Genomic findings other than those listed in Table 1 are not prescriptive or conclusive for labeled use of any specific therapeutic product.

Table 1. MI Cancer Seek Companion Diagnostic Indications

INDICATION	BIOMARKER	THERAPY
Breast Cancer	*PIK3CA* (C420R, E542K, E545A, E545D [1635G>T only], E545G, E545K, Q546E, Q546R, H1047L, H1047R, H1047Y)	PIQRAY® (alpelisib)
Colorectal Cancer (CRC)	*KRAS* wild-type (absence of mutations in exons 2, 3, and 4) and *NRAS* wild-type (absence of mutations in exons 2, 3, and 4)	VECTIBIX® (panitumumab)
	BRAF V600E	BRAFTOVI® (encorafenib) in combination with ERBITUX® (cetuximab)
Melanoma	*BRAF* V600E	BRAF Inhibitors approved by FDA*
	BRAF V600E or *BRAF* V600K	MEKINIST® (trametinib) or BRAF/MEK Inhibitor Combinations approved by FDA*
Non-small cell lung cancer (NSCLC)	*EGFR* exon 19 deletions and exon 21 L858R alterations	EGFR Tyrosine Kinase Inhibitors approved by FDA*
Solid Tumors	MSI-H	KEYTRUDA® (pembrolizumab), JEMPERLI (dostarlimab-gxly)
Endometrial Carcinoma	Not MSI-H	KEYTRUDA® (pembrolizumab) in combination with LENVIMA® (lenvatinib)

*For the most current information about the therapeutic products in this group, go to:
https://www.fda.gov/medical-devices/in-vitro-diagnostics/list-cleared-or-approved-companion-diagnostic-devices-in-vitro-and-imaging-tools#Group_Labeling

MI Cancer Seek™ is a single-site assay performed at Caris Life Sciences, Phoenix, AZ.

Contraindications:

There are no known contraindications.

Warnings and Precautions:

Biopsy may pose a risk to the patient when archival tissue is not available for use with the assay. The patient's physician should determine whether the patient is a candidate for biopsy.

Limitations:

- For *in vitro* diagnostic use.
- CAUTION: Federal law restricts this device to sale by or on the order of a physician.
- The acceptable preparation method for MI Cancer Seek CDx specimens is FFPE. Other preparations have not been evaluated.
- The test is designed to report out somatic variants and is not intended to report germline variants.
- MI Cancer Seek requires a minimum tumor percentage of 20% for detection of alterations, with tumor content enrichment recommended for specimens with tumor percentage lower than 20%.
- Genomic findings other than those listed in the Companion Diagnostic Indications table are not prescriptive or conclusive for labeled use of any specific therapeutic product. Confirmation of tumor mutation status using an FDA-approved CDx test is needed for therapeutic use.
- A negative result does not rule out the presence of a mutation below the limits of detection of the assay.
- MI Cancer Seek is only approved for use with Caris Life Sciences pre-qualified Illumina NovaSeq 6000 instruments.
- The test is intended to be performed on specific serial number-controlled instruments by Caris Life Sciences.
- MI Cancer Seek does not report TMB for values lower than 3 mut/Mb as the accuracy of TMB values below 3 mut/Mb are unreliable.
- Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all applicable information concerning the patient's condition, such as patient and family history, physical examinations, information from other diagnostic tests, and patient preferences, in accordance with the standard of care in a given community.
- Based on the low positive percent agreement (PPA) in the accuracy study for ERBB2 copy number amplifications (CNAs) in breast cancer patients, this alteration may not be detected. Additional clinical investigation to confirm the presence of ERBB2 CNAs in the breast cancer patient's tumor with another FDA approved or cleared test is strongly recommended.
- Patients with breast cancer whose specimens have intermediate ERBB2 CNA status should be tested with another FDA approved or cleared test to ascertain ERRB2 CNA status in their tumor.
- Test may have reduced sensitivity and may yield false negative results in samples where necrotic tissue is >15%, melanin is >5%, fat cells are >10%.

Test Principle:

MI Cancer Seek is a single-site assay performed at Caris Life Sciences located at 4610 South 44th Place, Phoenix, AZ 85040. The test includes reagents, software, and procedures for testing of total nucleic acid (TNA) from formalin fixed paraffin embedded (FFPE) tumor tissue. The test uses a custom bait panel to measure all coding regions of the exome to detect and report SNVs and indels within 228 genes outlined in Table 2 across solid tumors and amplifications in ERBB2 in patients with breast cancer only. The test also detects MSI (determined from 3,210 genes) and whole exome based TMB in patients with solid tumors.

Table 2. MI Cancer Seek Reportable Gene List for SNVs and indels

ABL1	*BCL9*	*CDKN2A*	*ESR1*	*FLT1*	*JAK3*	*MET*	*NRAS*	*PPP2R2A*	*RUNX1*	*STK11*
ACVR1	*BCOR*	*CHEK1*	*EZH2*	*FLT3*	*KDM5C*	*MITF*	*NSD1*	*PRDM1*	*SDHA*	*SUFU*
AIP	*BLM*	*CHEK2*	*FANCA*	*FOXA1*	*KDM6A*	*MLH1*	*NSD2*	*PRKACA*	*SDHAF2*	*TCF7L2*
AKT1	*BMPR1A*	*CIC*	*FANCB*	*FOXL2*	*KDR*	*MLH3*	*NTHL1*	*PRKAR1A*	*SDHB*	*TERT*
AKT2	*BRAF*	*CREBBP*	*FANCC*	*FUBP1*	*KEAP1*	*MPL*	*NTRK1*	*PRKDC*	*SDHC*	*TET2*
AKT3	*BRCA1*	*CSF1R*	*FANCD2*	*GATA3*	*KIT*	*MRE11*	*NTRK2*	*PTCH1*	*SDHD*	*TMEM127*
ALK	*BRCA2*	*CTCF*	*FANCE*	*GNA11*	*KLF4*	*MSH2*	*NTRK3*	*PTEN*	*SETD2*	*TNFAIP3*
AMER1	*BRIP1*	*CTNNA1*	*FANCF*	*GNA13*	*KMT2A*	*MSH3*	*PALB2*	*PTPN11*	*SF3B1*	*TNFRSF14*
APC	*BTK*	*CTNNB1*	*FANCG*	*GNAQ*	*KMT2C*	*MSH6*	*PBRM1*	*RAC1*	*SMAD2*	*TP53*
AR	*CALR*	*CXCR4*	*FANCI*	*GNAS*	*KMT2D*	*MTOR*	*PDGFRA*	*RAD50*	*SMAD4*	*TRAF7*
ARAF	*CARD11*	*CYLD*	*FANCL*	*H3F3A*	*KRAS*	*MUTYH*	*PDGFRB*	*RAD51B*	*SMARCA4*	*TSC1*
ARID1A	*CBFB*	*DDR2*	*FANCM*	*H3F3B*	*LZTR1*	*MYC*	*PIK3CA*	*RAD51C*	*SMARCB1*	*TSC2*
ARID2	*CCND1*	*DICER1*	*FAS*	*HIST1H3B*	*MAP2K1*	*MYCN*	*PIK3CB*	*RAD51D*	*SMARCE1*	*U2AF1*
ASXL1	*CCND2*	*DNMT3A*	*FAT1*	*HNF1A*	*MAP2K2*	*MYD88*	*PIK3R1*	*RAD54L*	*SMO*	*VHL*
ATM	*CCND3*	*EGFR*	*FBXW7*	*HOXB13*	*MAP2K4*	*NBN*	*PIK3R2*	*RAF1*	*SOCS1*	*WRN*
ATRX	*CD79B*	*EP300*	*FGFR1*	*HRAS*	*MAP3K1*	*NF1*	*PIM1*	*RASA1*	*SOS1*	*WT1*
AXIN2	*CDC73*	*EPHA2*	*FGFR2*	*IDH1*	*MAPK1*	*NF2*	*PMS2*	*RB1*	*SPEN*	*XPO1*
B2M	*CDH1*	*ERBB2*	*FGFR3*	*IDH2*	*MAX*	*NFE2L2*	*POLD1*	*RET*	*SPOP*	*XRCC1*
BAP1	*CDK12*	*ERBB3*	*FGFR4*	*IRF4*	*MED12*	*NFKBIA*	*POLE*	*RHOA*	*SRC*	
BARD1	*CDK4*	*ERBB4*	*FH*	*JAK1*	*MEF2B*	*NOTCH1*	*POT1*	*RNF43*	*STAG2*	
BCL2	*CDKN1B*	*ERCC2*	*FLCN*	*JAK2*	*MEN1*	*NPM1*	*PPP2R1A*	*ROS1*	*STAT3*	

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FDA Evidence Levels:

Genomic findings other than those listed in the Intended Use are not prescriptive or conclusive for labeled use of any specific therapeutic product. Test results should be interpreted in the context of pathological evaluation of tumors, treatment history, clinical findings, and other laboratory data. The test report includes genomic findings reported in the following levels (Table 3).

Table 3. Classification Criteria for FDA Evidence Levels

LEVEL	CRITERIA
Level 1	Biomarker is FDA-approved as a companion diagnostic as part of MI Cancer Seek™
Level 2	Cancer alterations that are well-established with strong clinical evidence that the clinician must know according to professional consensus guidelines in the specific tumor type.
Level 3	Cancer alterations with potential clinical significance, e.g., biomarkers deemed useful for directing patients to a clinical trial or simply for informational purposes.
	i. Clinical data such as case reports, single or several case series, or Phase I/II clinical trial data that support the utility of specific biomarker alteration to direct a patient to clinical trials, or ii. Pre-clinical and/or *in vitro* studies provide structural analysis of the mutation, fusion, or isoform to confirm pathogenicity (tumor-promoting), sensitivity, or resistance through colony forming assays, growth inhibition or drug sensitivity assays, etc.

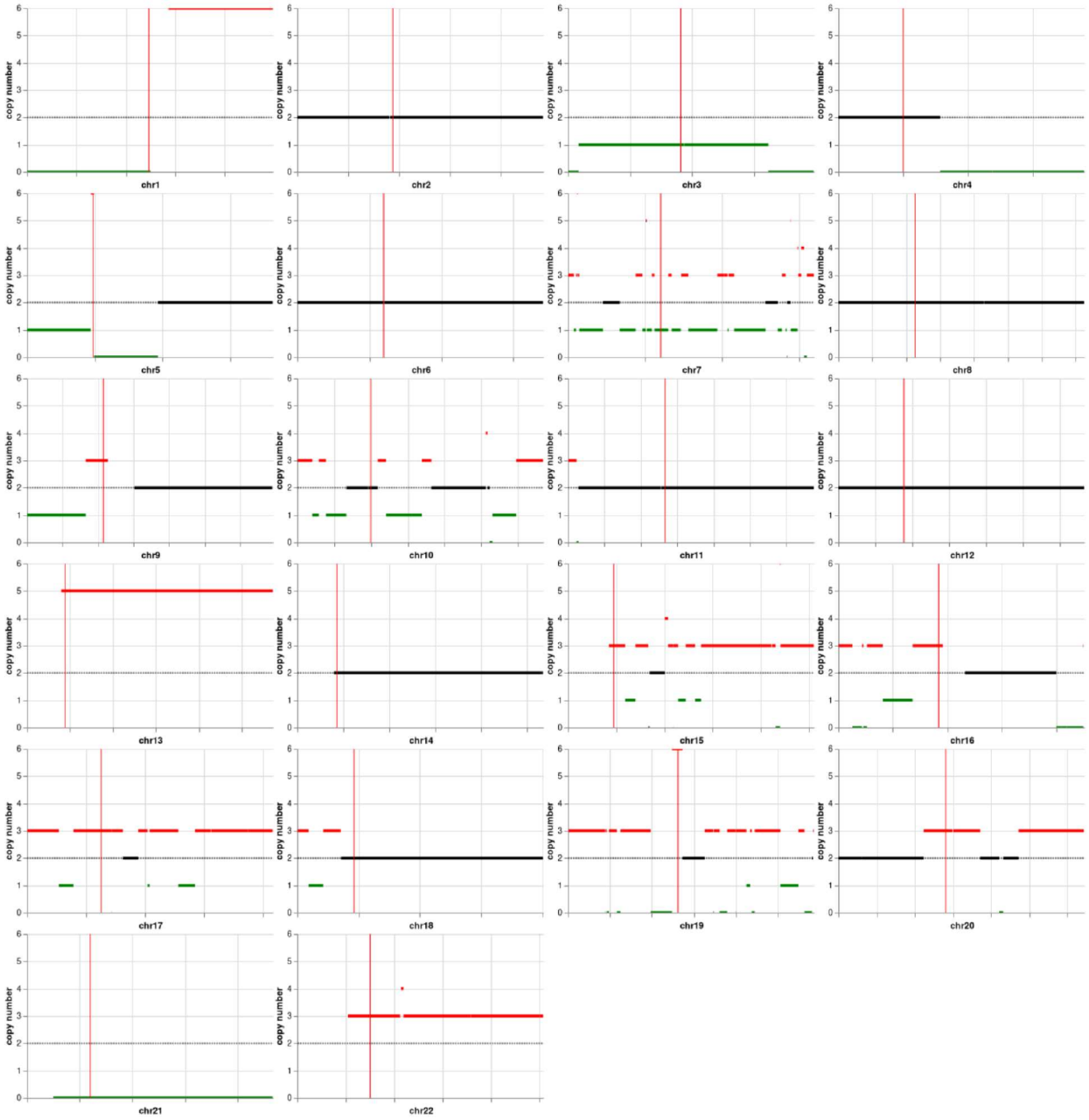
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Gene Expression

Gene	Percentile in Cancer Type	Gene	Percentile in Cancer Type	Gene	Percentile in Cancer Type
ADORA2A	90	IGF1R	6	PDCD1LG2	78
ALK	87	KDM1A	2	PIK3CA	32
ATM	80	KDR	65	PRAME	92
BRAF	82	KRAS	60	PTEN	70
BRCA1	58	LAG3	91	RB1	95
BRCA2	36	MAGEA4	67	RET	88
BRD4	74	MDM2	86	ROR1	75
CCND1	88	MET	29	ROR2	72
CCNE1	32	MKI67	21	ROS1	58
CCNG1	31	MSLN	8	SMAD4	99
CD274	72	MTAP	81	SRC	48
CD276	41	MTOR	2	SSTR2	66
CDKN2A	64	MUC1	48	SSTR3	87
CEACAM5	21	MUC16	23	SSTR5	92
CLDN18	57	MYC	26	TACSTD2	4
CLDN6	90	NECTIN4	94	TGFB1	47
CT83	46	NF1	68	TOP1	76
CTLA4	75	NRAS	13	TP53	92
EGFR	45	NRG1	86	TP53BP1	57
EPHA2	12	NTRK1	94	TSC1	88
ERBB2	22	NTRK2	68	TSC2	86
ERBB3	94	NTRK3	81	VEGFA	99
FGFR2	68	PALB2	72	XPO1	45
FGFR3	98	PARP1	58	ZG16B	91
FOLR1	66	PARP2	76		
HRAS	50	PDCD1	86		

Gene Expression of Selected Genes by Whole Transcriptome Sequencing (WTS) Methods:
Gene expression is derived from whole transcriptome sequencing. Relative expression of genes are calculated as normalized values using Transcripts per Million Molecules or TPM. TPM is presented as a percentile derived by comparison to a distribution of Caris' internal cohort of the tumor-type profiled. Selected genes reported in this section were chosen based on their tumor-type specific relevance for matching to clinical trials, or tumor type subclassification.

Karyotype



Karyotyping using Copy Number Analysis by Whole Exome Sequencing (WES) Methods:
Whole exome sequencing in combination with interrogation of single nucleotide polymorphisms (SNPs) tiled throughout the genome, allows for the identification and visualization of cytogenetic aberrations.

Somatic structural variants like whole or partial chromosome duplications or deletions, are important for cancer development and progression, and may identify clinically actionable alterations.

Mutational Analysis by Next-Generation Sequencing (NGS)

TUMOR MUTATIONAL BURDEN	
Mutations / Megabase	**Result**
3	Low

TMB

Tumor Mutational Burden (TMB) is defined as the number of somatic non-synonymous mutations per million bases of sequenced DNA in a tumor sample. Tumors with high TMB may increase the number of neoantigens which is hypothesized to increase T-cell reactivity and potential for response to immune checkpoint inhibitors. TMB analysis was performed based on next generation sequencing analysis of genomic DNA isolated from a tumor sample.

MICROSATELLITE INSTABILITY ANALYSIS	
Test	**Result**
MSI	Stable

MSI

Microsatellite instability (MSI) status is a measure of the number of somatic mutations within short, repeated sequences of DNA (microsatellites). MSI-High status can indicate that the tumor has a defect in mismatch repair (MMR) abrogating the ability to correct mistakes during DNA replication. Tumors with MSI-high status may increase the number of neoantigens which is hypothesized to increase T-cell reactivity and potential for response to immune checkpoint inhibitors. Tumor-only microsatellite instability status by NGS (MSI-NGS) is measured by the direct analysis of known microsatellite regions sequenced in the CMI NGS panel.

GENOMIC LOSS OF HETEROZYGOSITY	
Test	**Result**
Genomic Loss of Heterozygosity (LOH)	Low - 8% of tested genomic segments exhibited LOH (assay threshold is ≥ 16%)

LOH

To calculate genomic loss-of-heterozygosity (LOH), the 22 autosomal chromosomes are split into 552 segments and the LOH of single nucleotide polymorphisms (SNPs) within each segment is calculated. Caris WES data consist of approximately 250k SNPs spread across the genome. SNP alleles with frequencies skewed towards 0 or 100% indicate LOH (heterozygous SNP alleles have a frequency of 50%). The final call of genomic LOH is based on the percentage of all 552 segments with observed LOH.

Additional Next-Generation Sequencing results continued on the next page. >

Mutational Analysis by Next-Generation Sequencing (NGS)

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
TERT	DNA-Tumor	Pathogenic Variant	-	0	c.-124C>T	33	NM_198253.2

Interpretation: A pathogenic variant was detected in TERT promoter region, also referred to as C228T in hg19.

Telomerase is a ribonucleoprotein polymerase that maintains telomere ends by addition of the telomere repeat TTAGGG. Telomerase expression plays a role in cellular senescence, as it is normally repressed in postnatal somatic cells resulting in progressive shortening of telomeres. Deregulation of telomerase expression in somatic cells may be involved in oncogenesis. Studies in mouse suggest that telomerase also participates in chromosomal repair, since de novo synthesis of telomere repeats may occur at double-stranded breaks.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
TGFBR2	DNA-Tumor	Pathogenic Variant	p.N490fs	6	c.1469dupA	38	NM_003242.5

Interpretation: A pathogenic variant was detected in TGFBR2.

This gene encodes a member of the Ser/Thr protein kinase family and the TGFB receptor subfamily. The encoded protein is a transmembrane protein that has a protein kinase domain, forms a heterodimeric complex with another receptor protein, and binds TGF-beta. This receptor/ligand complex phosphorylates proteins, which then enter the nucleus and regulate the transcription of a subset of genes related to cell proliferation. Mutations in this gene have been associated with Marfan Syndrome, Loeys-Deitz Aortic Aneurysm Syndrome, and the development of various types of tumors.

Additional Next-Generation Sequencing results continued on the next page. >

Mutational Analysis by Next-Generation Sequencing (NGS)

GENES TESTED WITH INDETERMINATE* RESULTS BY TUMOR DNA SEQUENCING					
ATP6AP2	CUL3	PLCB4	PRKD1	REST	STAG2
COL2A1	KIF1B	PRDM6	RASA1	SMARCA2	

*** Genes in this table were ruled indeterminate due to low coverage for some or all exons.**

For a complete list of genes tested, visit www.CarisMolecularIntelligence.com/profilemenu.

NGS Methods

Direct sequence analysis was performed on genomic DNA isolated from a micro-dissected tumor sample using Illumina NovaSeq 6000 sequencers. A hybrid pull-down panel of baits was used to enrich more than 700 clinically relevant genes along with > 20,000 other genes. Sequence data is analyzed using a customized bioinformatics pipeline to detect sequencing variants, copy number alterations (amplifications and deletions) indels and HLA genotypes. In addition, genomic signatures for tumor mutational burden (TMB), microsatellite instability (MSI), genomic loss-of-heterozygosity (LOH) or HRD-Genomic Scar Score (HRD-GSS), and homologous recombination deficiency (HRD) are reported when applicable. For a complete list of what is covered by the assay, and genes with partial coverage, please contact Caris Customer Support.

Copy Number Alterations by Next-Generation Sequencing (NGS)

GENES TESTED WITH INTERMEDIATE CNA RESULTS					
AKT3	DDR2	MCL1	NTRK1	RAC1	

GENES WITH INDETERMINATE CNA RESULTS					
AFF1	ERCC8	JAK1	MTOR	RUNX1	TNFRSF14
ARID1A	ERG	KLHL6	NOTCH2	SDHB	U2AF1
ATP1A1	ETV5	LCK	PIK3CA	SPEN	
BCL6	FAT1	LPP	PIK3R1	STIL	
CAMTA1	FBXW7	MAP3K1	RAD54L	TET2	
CDKN2C	FUBP1	MDS2	RPL22	TFRC	

CNA Methods

The copy number alteration (CNA) of each exon is determined by a calculation using the average sequencing depth of the sample along with the sequencing depth of each exon and comparing this calculated result to a pre-calibrated value. A complete list of genes for reporting copy number alterations, including amplifications and deletions, is available upon request.

Gene Fusion and Transcript Variant Detection by RNA Sequencing

GENES TESTED WITH GENE FUSION OR TRANSCRIPT VARIANT DETECTED				
Biomarker	**Fusion/Isoform**	**Splice Site**	**Transcript ID**	**Variant Interpretation**
NRG1	CDH1:NRG1	exon 10:exon 6	NM_001317184.2/NM_001159999.3	Pathogenic Fusion

Interpretation: A CDH1::NRG1 fusion was detected by RNA sequencing. This fusion is in-frame and similar to other reported fusions involving NRG1 (Dhanasekaran 2014 Nat Commun 5:5893; Fernandez-Cuesta 2014 Cancer Discov 4:415). Exon 10 of CDH1 (NM_001317184.2) is joined to exon 6 of NRG1 (NM_001159999.3).

NRG1 is a protein involved in the ERBB2/ERBB3 - PI3K/AKT pathway. Fusions and other alterations in NRG1 may activate this pathway. While rare across all cancer types, NRG1 fusions are found most commonly in mucinous adenocarcinomas of the lung.

Whole Transcriptome Sequencing (WTS) Methods

Gene fusion and variant transcript detection were performed on RNA isolated from a tumor sample using next generation sequencing. The assay also detects fusions occurring at known and novel breakpoints within genes. The genes included in this report represent the subset of genes associated with cancer. The complete list of unclassified alterations is available by request.

Protein Expression by Immunohistochemistry (IHC)

Biomarker	Patient Tumor			Thresholds
	Staining Intensity (0, 1+, 2+, 3+)	Percent of cells	Result	Conditions for a Positive Result:
PD-L1 (SP142)	0	100	Negative	Intensity ≥2+ and ≥5% of cells stained

Her-2 IHC: Biopsy	
Final Score	Threshold for Positive Result
Negative | Score 0	Intensity =3+ and ≥5 tumor cells staining

Clones used: ERBB2 (Her2/Neu) (4B5), PD-L1 (SP142).

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Electronic Signature
[redacted]
02/19/2025

IHC Methods

The Laboratory Developed Tests (LDT) immunohistochemistry (IHC) assays were developed and their performance characteristics determined by Caris Life Sciences. These tests have not been cleared or approved by the US Food and Drug Administration. The FDA has determined that such clearance or approval is not currently necessary. Interpretations of all immunohistochemistry (IHC) assays were performed manually or with the assistance of an AI-based image analysis tool by a board certified pathologist using a microscope and/or digital whole slide image(s).

The following IHC assays were performed using FDA-approved companion diagnostic or FDA-cleared tests consistent with the manufacturer's instructions: ALK (VENTANA ALK (D5F3) CDx Assay, Ventana), ER (CONFIRM anti-Estrogen Receptor (ER) (SP1), Ventana), FOLR1 (VENTANA FOLR1-2.1 RxDx, Ventana), CLDN18 (VENTANA, 43-14A RxDx Assay), PR (CONFIRM anti-Progesterone Receptor (PR) (1E2), Ventana), HER2/neu (PATHWAY anti-HER-2/neu (4B5), Ventana), Ki-67 (MIB-1 pharmaDx, Dako), MAGE-A4 1F9 (pharmDx, Dako), PD-L1 22c3 (pharmDx, Dako), PD-L1 SP142 (VENTANA, non-small cell lung cancer), PD-L1 28-8 (pharmDx, Dako, gastric / GEJ, non-small cell lung cancer), PD-L1 SP263 (Ventana, non-small cell lung cancer), and Mismatch Repair (MMR) proteins (MLH1, MSH2, MSH6, and PMS2; VENTANA MMR RxDx Panel, Ventana).

HER2 results and interpretation follow the ASCO/CAP scoring criteria. Bartley, A.N., J.A. Ajani, et al. (2016). "HER2 testing and clinical decision making in gastroesophageal adenocarcinoma: guideline from the College of American Pathologists, American Society for Clinical Pathology, and the American Society of Clinical Oncology". J Clin Oncol. 35(4):446-464.

Professional Component Performed: - - Location 67, Chapel Hill, NC. [redacted] Lab Director.

References

#	Drug	Biomarker	Reference
1	zenocutuzumab	NRG1	Schram, A., E.M. O'Reilly, et al (2023). "Durable efficacy of zenocutuzumab, a HER2 x HER3 bispecific antibody, in advanced NRG1 fusion positive (NRG1+) pancreatic ductal adenocarcinoma (PDAC)." AnnOncol 34(Suppl 2): S895-S896. View Citation Online
2	zenocutuzumab	NRG1	Schram, A., R. Somwar, et al. (2022). "Zenocutuzumab, a HER2xHER3 Bispecific Antibody, Is Effective Therapy for Tumors Driven by NRG1 Gene Rearrangements." Cancer Discovery. 12 (5): 1233-1247. View Citation Online

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